AP PRECALCULUS · TOPIC 1.13

Function Model Selection and Assumption Articulation

Section 1 — Multiple Choice
Circle the letter of the correct answer **SOLVED**

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 A table gives $y = 5, 9, 13, 17, 21$ for $x = 0, 1, 2, 3, 4$. Which model fits the data?

A. Linear, $y = 4x + 5$
B. Quadratic, $y = 4x^2 + 5$
C. Linear, $y = 5x + 4$
D. Linear, $y = 9x + 5$

A. Every first difference is 4, so the rate is constant: slope 4, intercept 5. B: assumed increasing data is quadratic. C: swapped slope and intercept. D: used $y(1)$ as the slope.

2 A table gives $y = 3, 4, 9, 18, 31$ for $x = 0, 1, 2, 3, 4$. Which function fits the data?

A. $y = 4x^2 - x + 3$
B. $y = 2x^2 + x + 3$
C. $y = 2x^2 - x + 3$
D. $y = 2x^2 - x + 4$

C. First differences $1, 5, 9, 13$; second differences $4, 4, 4$, so $2a = 4$ and $a = 2$; then $c = 3$ and $b = -1$. A: used the second difference as a. B: sign error in b. D: took c from $y(1)$.

3 A table gives $y = 2, 3, 10, 29, 66, 127$ for $x = 0, 1, 2, 3, 4, 5$. The nth differences are constant. What is n?

A. 2
B. 3
C. 5
D. 6

B. Differences: $1, 7, 19, 37, 61$; then $6, 12, 18, 24$; then $6, 6, 6$. The third differences are constant. A: stopped at the second differences. C: used the number of points minus one. D: gave the value of the constant difference.

4 A rectangle has perimeter 60 m and width w. Which function type best models its area $A(w)$?

A. Linear
B. Cubic
C. Piecewise-defined
D. Quadratic

D. Length is $30 - w$, so $A(w) = w(30 - w)$, a two-dimensional (area) context. A: assumed linear because the perimeter is linear in w. B: confused area with volume. C: no change of rule over intervals occurs.

5 A rectangle has perimeter 60 m and width w, so its area is $A(w) = w(30 - w)$. Which domain restriction fits the context?

A. All real numbers
B. $w > 0$
C. $0 < w < 30$
D. $0 < w < 60$

C. Width and length $30 - w$ must both be positive. A: no restriction considered. B: only checked the width. D: used the perimeter, 60, as the upper limit.

6 Eight points on a graph have distinct input values. A polynomial function of degree at most n passes through all eight points. What is the least value of n that is guaranteed?

A. 8
B. 7
C. 6
D. 4

B. $n+1$ points with distinct inputs lie on a polynomial of degree at most n, so $n = 7$. A: set n equal to the number of points. C: subtracted one too many. D: halved the number of points.

7 Which situation is best modeled by a piecewise-defined function?

A. A taxi fare of 3 dollars for the first 2 km, then 1.50 dollars for each additional km
B. The distance driven at a constant 60 km/h
C. The area of a square of side s
D. The volume of a cube of edge s

A. The rate of change differs on two intervals. B: constant rate gives a linear model. C: area gives a quadratic. D: volume gives a cubic.

8 A model for tickets sold is $T(p) = 200 - 8p$, where p is the ticket price in dollars. Which assumption is built into the model?

A. Sales stay at 200 tickets whatever the price

B. Sales increase as the price increases

C. Prices must be whole numbers of dollars

D. Each 1 dollar increase in price lowers sales by 8 tickets

D. The model is linear: the rate of change is constant at -8 tickets per dollar. A: read the intercept as a constant. B: sign error on the slope. C: confused a domain restriction with an assumption about change.

AP PRECALCULUS · TOPIC 1.13 · SECTION 2

Function Model Selection and Assumption Articulation

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

The number of visitors, in hundreds, to a museum exhibit on day x after it opens is recorded: $x = 0, 1, 2, 3, 4, 5$ gives $y = 12, 19, 24, 27, 28, 27$.

(a) Show that a quadratic model is appropriate for these data. DOK3 2 pt

First differences $7, 5, 3, 1, -1$; second differences $-2, -2, -2, -2$ are constant, so the model is quadratic.

(b) Find a quadratic model and use it to find the greatest number of visitors and the day it occurs. DOK3 2 pt

$a = -1$ (since $2a = -2$), then $y = -x^2 + 8x + 12$. Vertex $x = 4$, $y = 28$: 2800 visitors on day 4.

(c) The model is used until $y = 0$. Find that value of x to three decimal places and state one reason the model should not be used beyond it. DOK3 2 pt

$x^2 - 8x - 12 = 0 \Rightarrow x = 4 + \sqrt{28} \approx 9.292$. After this, $y < 0$, a negative number of visitors, which is impossible (domain restriction).

Question 10 No calculator · 6 points

An open box is made from an 18 cm by 18 cm square sheet by cutting a square of side x cm from each corner and folding up the sides, so $V(x) = x(18 - 2x)^2$.

(a) State a domain for V that fits the context and give the reason for each end. DOK3 2 pt

$0 < x < 9$: the height x must be positive and the base side $18 - 2x$ must be positive.

(b) State the type and degree of V and justify the choice using the context. DOK3 2 pt

Cubic, degree 3: a volume (three dimensions) and the product of three linear factors.

(c) A different data set has $(0, 1)$, $(1, 2)$, $(2, 5)$, $(3, 10)$. Use differences to find the type, then find y when $x = 5$. DOK3 2 pt

First differences $1, 3, 5$; second differences $2, 2$: quadratic, $y = x^2 + 1$. At $x = 5$: $y = 26$.

Answer key (Section 1): 1A 2C 3B 4D 5C 6B 7A 8D